Unit 5.2 – Pharmacologic and Parenteral Therapies (Fluids, Electrolytes, and Key Drug Classes)

Lilley Chapter 30 – Fluid and Electrolytes (Full Chapter)

Core Fluid Concepts and Compartments

- Total body water (TBW) is about 60% of adult body weight (higher in infants, lower in older adults), distributed between intracellular fluid (ICF) and extracellular fluid (ECF).[^1]
- ECF is subdivided into intravascular fluid (IVF/plasma volume) and interstitial fluid (ISF); transcellular fluid (e.g., CSF, synovial, pleural) is a specialized part of ISF.[^1]
- Normal fluid balance (euvolemia) is maintained via thirst, antidiuretic hormone (ADH), aldosterone, and renal function; movement between compartments occurs through diffusion, osmosis, active transport, and filtration.[^1]

Osmolality, Tonicity, and Pressures

- **Osmolality** refers to concentration of particles in body fluids (normal ~290–310 mOsm/kg), while **tonicity** describes the effective osmolality of IV solutions relative to body fluids.[^1]
- **Isotonic** solutions have similar osmolality to body fluids; **hypotonic** solutions have lower osmolality (fluid moves into cells), and **hypertonic** solutions have higher osmolality (fluid moves out of cells).[^1]
- Colloid oncotic pressure (COP) is the pulling pressure exerted by plasma proteins (primarily albumin) that draws fluid into the intravascular space (~24 mm Hg), opposed by interstitial hydrostatic pressure (~17 mm Hg).[^1]

Dehydration and Types of Fluid Loss

- Dehydration is excessive water loss, often with associated electrolyte imbalance (notably sodium, potassium, chloride); it can be **hypertonic**, **hypotonic**, or **isotonic** depending on relative water vs sodium loss.[^1]
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  - Hypertonic dehydration: water loss > sodium loss (e.g., fever, sweating) → cells shrink.[^1]
  - Hypotonic dehydration: sodium loss > water loss (e.g., kidney insufficiency, low aldosterone) → fluid shifts into cells.[^1]
  - Isotonic dehydration: equal sodium and water loss (e.g., vomiting, diarrhea) → ECF volume deficit.[^1]

- Clinical conditions causing fluid loss include bleeding, bowel obstruction, diarrhea, fever, and vomiting; common symptoms are tachycardia, hypotension, decreased urine output, dry mucous membranes, poor skin turgor, and weight loss.[^1]

Crystalloids

- Crystalloids are IV fluids containing water and small solutes that diffuse across membranes; they expand ECF volume and correct specific fluid and electrolyte disturbances.[^1]
- Common crystalloids and tonicity:[^1]
  - 0.9% NaCl (NS): isotonic (308 mOsm), primary resuscitation and maintenance fluid.
  - Lactated Ringer's: isotonic balanced electrolyte solution (Na, K, Ca, lactate).
  - 3% NaCl: hypertonic (for severe hyponatremia—high-risk, slow infusion, frequent sodium checks).
  - 0.45% NaCl: hypotonic (shifts fluid into cells; risk of hemolysis if given alone).
  - D5W: isotonic in the bag but acts hypotonic once dextrose is metabolized.
- Indications: maintenance fluids, replacement of fluid losses, promoting urine output; better for dehydration than pure plasma volume expansion.[^1]
- Adverse effects: edema (peripheral, pulmonary), dilution of plasma proteins and COP, short duration of intravascular effect, acid–base shifts with large volumes; contraindicated in hypervolemia and some electrolyte disturbances.[^1]
- Nursing: choose solution by tonicity and patient status; monitor for fluid overload (lungs, edema, weight, JVP), electrolyte changes, compatibility (e.g., some drugs precipitate in LR or dextrose).[^1]

Colloids

- Colloids are large-molecule solutions (e.g., albumin, dextran, hetastarch, pentastarch) that stay in the vasculature and increase COP, pulling fluid from ISF into the intravascular space; used as plasma expanders.[^1]
- Indications: shock, burns, hypoproteinemia, and other conditions needing sustained plasma volume expansion.[^1]
- Advantages: maintain plasma volume longer than crystalloids, require smaller volumes to expand intravascular volume.[^1]
- Disadvantages/adverse effects: no oxygen-carrying capacity or clotting factors; dilutional coagulopathy and bleeding risk; expensive; rare anaphylaxis (dextran) and renal issues; contraindicated in hypervolemia, severe anemia, heart failure, and some electrolyte disorders.[^1]
- Nursing: infuse slowly; monitor for signs of fluid overload (SOB, crackles, increased weight, edema); check Hgb/Hct and total protein; albumin should be room temperature; adhere to institutional criteria for albumin use.[^1]

Blood Products

- Blood products (packed RBCs, whole blood, fresh frozen plasma, cryoprecipitate, platelets) are biological drugs used when more than ~25% of blood volume is lost or when oxygen-carrying or clotting capacity must be restored.[^1]
- Mechanism: increase COP and plasma volume like colloids; RBC-containing products also restore oxygen-carrying capacity.[^1]
- Indications:[^1]
  - PRBCs: symptomatic anemia, Hgb deficits, 20–25% blood loss.
  - Whole blood: major hemorrhage (>25% loss) needing volume plus clotting factors.
  - FFP: clotting factor deficiencies and as adjunct in massive transfusion.
  - Cryoprecipitate and plasma protein fraction: acute bleeding and specific factor deficiencies.
- Risks: transfusion reactions, anaphylaxis, transmission of infections (greatly reduced but never zero), volume overload, and dilutional coagulopathy.[^1]
- Nursing: verify type and crossmatch, follow two-RN check policy, use only 0.9% NaCl for blood; monitor VS before, during, after; stop transfusion and follow institutional protocol if any reaction suspected.[^1]

Electrolyte Physiology and Replacement – Potassium

- Potassium is the major intracellular cation (ICF ~150 mmol/L; serum 3.5–5 mmol/L); vital for nerve impulse transmission, muscle contraction (especially cardiac), acid–base balance, and cellular metabolism.[^1]
- **Hyperkalemia** (K >5–5.5 mmol/L): causes include kidney failure, ACE inhibitors, potassium-sparing diuretics, metabolic acidosis, tissue breakdown, burns, and excessive K intake.[^1]
- **Hypokalemia** (K <3.5 mmol/L): often due to GI loss, diuretics, corticosteroids, alkalosis, hyperaldosteronism, laxative misuse, and malabsorption.[^1]
- Early hypokalemia signs: anorexia, hypotension, lethargy, confusion, nausea, muscle weakness; late signs: dysrhythmias, neuropathy, paralytic ileus, secondary alkalosis.[^1]
- Treatment: identify and treat cause; oral dietary K for mild cases; oral or IV KCl or other salts for significant deficits.[^1]
- IV potassium: **high-alert**; must be diluted; maximum 10 mmol/hr via peripheral line and usually ≤20 mmol/hr via central line with cardiac monitoring; never IV push; monitor site for phlebitis and infiltration; ensure adequate urine output.[^1]
- Hyperkalemia treatment: IV bicarbonate, insulin with dextrose, calcium salts to stabilize myocardium, sodium polystyrene sulfonate (Kayexalate), and/or hemodialysis; restrict K intake.[^1]

Electrolyte Physiology and Replacement – Sodium

- Sodium is the main extracellular cation (135–145 mmol/L); regulates water distribution, ECF volume, osmotic pressure, and participates in acid–base balance.[^1]
- **Hyponatremia** (Na <135 mmol/L): causes include GI loss, excessive sweat, kidney disorders, SIADH, diuretics, and excess water intake; symptoms include lethargy, hypotension, cramps, vomiting, diarrhea, and seizures.[^1]
- **Hypernatremia** (Na >145 mmol/L): commonly due to dehydration or poor renal excretion; signs include red flushed skin, dry mucosa, intense thirst, temperature elevation, edema, hypertension, and decreased urine output.[^1]
- Mild hyponatremia: oral NaCl tablets and fluid restriction; more severe: IV NS or LR; severe symptomatic hyponatremia may require **carefully monitored** 3% NaCl (high-risk for osmotic demyelination if corrected too fast).[^1]
- Tolvaptan (Samsca), a vasopressin receptor antagonist, is used for euvolemic hyponatremia with close sodium monitoring and attention to CYP3A4 interactions.[^1]
- Nursing: monitor labs, hydration status, weight, neuro status; educate on safe salt tablet use and fluid intake; watch for signs of hypo- and hypernatremia; avoid rapid Na correction.[^1]

Nursing Process for Fluids and Electrolytes

- **Assessment**: medical history (cardiac, renal, hepatic, GI), medication and NHP use, dietary intake, I&O, vital signs, daily weights, skin turgor, mucous membranes, JVP, lung sounds, mental status, lab values (Na, K, Cl, creatinine, urea, Hct, Hgb).[^1]
- **Planning/Goals**: restore or maintain fluid and electrolyte balance, prevent falls and injury, avoid transfusion reactions and fluid overload.[^1]
- **Implementation**: verify orders; choose appropriate solution and route; monitor infusion rate and IV site (infiltration, phlebitis); titrate diuretics or fluids based on response; educate about signs/symptoms to report (dizziness, palpitations, SOB, edema).[^1]
- **Evaluation**: stable vital signs, improved lab values, adequate urine output (≥30 mL/hr), improved mental status, resolution of hypovolemia/overload signs, no serious adverse events.[^1]

Key Drug Classes – Nursing Knowledge, Administration, Assessment, Implications

> These are high-yield NCLEX/Canadian RNCP categories. Use your drug handbook and course notes for specific agents, doses, and local policies.

Anticoagulants, Coagulation Factor Inhibitors, Thrombolytics

- **Anticoagulants** (heparin, LMWHs, warfarin, DOACs like rivaroxaban, apixaban, dabigatran) prevent clot formation/extension; they do not dissolve existing clots.[^2]
- **Thrombolytics** (e.g., alteplase/tPA, tenecteplase) actively lyse clots by converting plasminogen to plasmin; used in acute MI, ischemic stroke, massive PE under strict criteria.[3][4]
- Major risk for all: **bleeding** (GI, intracranial, injection sites, hematuria, epistaxis, bruising); requires bleeding precautions and regular assessment.[4][3]
- Monitoring:
  - Heparin: aPTT and platelets (HIT risk).[3][2]
  - Warfarin: PT/INR; maintain consistent vitamin K intake.[2][3]
  - DOACs: no routine lab titration but renal function and bleeding assessment are essential.[^2]
- Nursing: double-check dosing (weight-based for heparin), avoid IM injections, use infusion pumps, educate about signs of bleeding, interactions (e.g., NSAIDs), and adherence.[3][2]

Cardiac Medications and Diuretics (High-Level)

- **Cardiac meds** include anti-hypertensives (ACEIs, ARBs, beta-blockers, CCBs), anti-anginals, anti-arrhythmics, and heart-failure drugs (e.g., digoxin, vasodilators); all require BP/HR and symptom monitoring.[5][6]
- **Diuretics** (loop, thiazide, potassium-sparing) reduce volume and cardiac workload; loop (furosemide) and thiazides risk hypokalemia and other electrolyte changes; potassium-sparing (spironolactone) risk hyperkalemia.[6][5]
- Nursing: monitor daily weights, I&O, renal function, electrolytes (Na, K), BP, orthostatic symptoms, hearing (loop ototoxicity), and teach about time of dosing (morning), K-rich vs low-K diet depending on agent.[7][5] [^6]

Antidiabetics and Insulins

- Insulin categories: rapid-acting, short-acting, intermediate (NPH), and long-acting (e.g., glargine, detemir); each has specific onset, peak, and duration that must be coordinated with meals to prevent hypoglycemia.[8][9]
- Oral agents (metformin, sulfonylureas, DPP-4 inhibitors, SGLT2 inhibitors, etc.) target insulin secretion, sensitivity, or glucose excretion.
- Insulin is a **high-alert medication**: confirm indication, check most recent blood glucose, assess for hypoglycemia, label syringes, perform independent double-checks per policy.[9][10]
- Nursing: rotate injection sites, monitor for hypoglycemia (shakiness, confusion, sweating, tachycardia), adjust dosing during NPO status, illness, or steroid use; provide teaching on self-monitoring, sick-day rules, and injection technique.[10][9]

Bronchodilators

- Three main bronchodilator classes: **beta$_2$-agonists** (e.g., salbutamol/albuterol, salmeterol), **anticholinergics** (ipratropium, tiotropium), and **methylxanthines** (theophylline).[11][12] [^13]
- Short-acting beta$_2$-agonists (SABAs) are rescue meds for acute bronchospasm; long-acting (LABAs) are maintenance, often combined with inhaled corticosteroids.[12][11]
- Anticholinergics are key in COPD; methylxanthines are less commonly used due to narrow therapeutic window and drug interactions.[13][11]
- Nursing: assess respiratory status pre/post (RR, effort, O_2 sat, peak flow), teach proper inhaler/neb technique and sequence (bronchodilator before steroid), monitor for tremor, tachycardia (beta$_2$-agonists), dry mouth/urinary retention (anticholinergics), and serum levels for theophylline.[11][12] [^13]

Opioids, Narcotics, Sedatives

- **Opioids** (e.g., morphine, hydromorphone, fentanyl, oxycodone) are potent analgesics; sedatives/hypnotics (e.g., benzodiazepines, z-drugs) depress CNS activity.
- Major risks: respiratory depression, excessive sedation, hypotension, constipation, nausea, urinary retention, tolerance, physical dependence, and misuse.[^14]
- Nursing: assess pain, sedation (e.g., sedation scale), RR and depth, O_2 sat; use lowest effective dose; avoid concurrent CNS depressants when possible; provide bowel regimen; keep naloxone or flumazenil available per protocol.[^14]

Antibiotics, Antifungals, Anti-infectives

- Antibiotics (e.g., beta-lactams, macrolides, fluoroquinolones), antifungals (e.g., azoles, amphotericin), and antivirals require organism-directed selection where possible and attention to resistance.
- Nursing: check allergies, obtain cultures before first dose when indicated, monitor for hypersensitivity, organ toxicity (renal, hepatic, ototoxicity), and superinfections; ensure completion of prescribed course and proper infusion rates.[^15]

Pain Medications, Anti-inflammatories, Corticosteroids

- Non-opioid analgesics (acetaminophen, NSAIDs) are first-line for mild–moderate pain; NSAIDs carry GI, renal, and CV risks.[^14]
- Corticosteroids have strong anti-inflammatory and immunosuppressive effects but cause many adverse effects (hyperglycemia, osteoporosis, infection risk, mood changes, fluid retention, adrenal suppression) with prolonged use.
- Nursing: use lowest effective dose for shortest duration, give with food (NSAIDs, oral steroids), protect GI tract when indicated, monitor glucose, BP, mood, and signs of infection; taper steroids rather than stop abruptly.[^14]

Medications for Mental Health Challenges (High-Level)

- Classes include antidepressants (SSRIs, SNRIs, TCAs, MAOIs), antipsychotics (typical and atypical), mood stabilizers (lithium, anticonvulsants), anxiolytics (benzodiazepines, buspirone), and sleep agents.
- Many carry black-box warnings (e.g., suicidality in youth for antidepressants) or serious adverse effects (e.g., metabolic syndrome and EPS with antipsychotics, lithium toxicity, serotonin syndrome).
- Nursing: baseline and ongoing mental-status assessments, suicide risk assessments, metabolic monitoring (weight, lipids, glucose), lab monitoring (e.g., lithium levels, CBC for certain antipsychotics), and patient teaching (adherence, onset of effect, interactions like tyramine with MAOIs).[^16]

Medications for Specific Circumstances

Epidural or Intrathecal Medications

- Used for regional analgesia (e.g., epidural opioids and local anesthetics for labour, post-op, or chronic pain), and for intrathecal chemotherapy or baclofen.
- Advantages: strong pain relief with lower systemic doses; risks: hypotension, motor block, urinary retention, respiratory depression (late with hydrophilic opioids), infection, catheter misplacement.
- Nursing: frequent neuro and respiratory assessments, monitor dermatomal level of block, maintain strict aseptic technique, label lines clearly (no IV meds into epidural), and manage pump safety; have protocols for treating hypotension and respiratory depression.[^14]

Medications Absolutely Contraindicated in Pregnancy (Conceptual)

- Teratogenic medications must be avoided unless benefits outweigh risks; examples in pharmacology texts commonly include certain **ACEIs/ARBs**, **warfarin**, some **anticonvulsants**, **methotrexate**, **isotretinoin**, and certain **hormonal agents** (e.g., some anti-neoplastics and strong androgens).[^15]
- Hormonal contraception such as depot medroxyprogesterone acetate (Depo-Provera) is **contraindicated in known or suspected pregnancy**; it is used only to prevent pregnancy.[^17]
- Nursing: always ask about pregnancy intentions and status, verify pregnancy testing when indicated, and provide clear education on teratogenic risks and contraception needs when prescribing high-risk drugs.[^17]

Injectable Depot Medications with Long-Acting Effects

- **Depot formulations** (IM/SC long-acting injections) exist for several drug classes: long-acting antipsychotics (e.g., depot risperidone, paliperidone, aripiprazole), contraceptives (e.g., depot medroxyprogesterone), and some opioids or hormonal agents.[18][16]
- Advantages: improved adherence, stable plasma levels, reduced daily pill burden; considerations: cannot be quickly reversed, so adverse effects and pregnancy exposure are harder to manage.[18][16]
- Nursing: verify correct drug, dose, and site; use Z-track technique when indicated; counsel about duration of action and delayed washout; assess side effects and monitor labs (e.g., metabolic profile for antipsychotics, bone density for long-term depot contraception).[16][18]

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